

Linear Algebra for AI

Final Exam 2024

① $T: \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be the map given by $T(x) = \begin{bmatrix} x_1 + 4x_2 + 5x_3 + 2x_4 \\ 2x_1 + x_2 + 3x_3 \\ -x_1 + 3x_2 + 2x_3 + 2x_4 \end{bmatrix}$.

a) Is the map linear? Why?

$T(x)$ can be given by $T(x) = Ax$, where $A = \begin{bmatrix} 1 & 4 & 5 & 2 \\ 2 & 1 & 3 & 0 \\ -1 & 3 & 2 & 2 \end{bmatrix}$ is the standard matrix of the map T . This means that T is a matrix map, and matrix maps are linear.
(The question already says that T is linear by mistake, so pointing that out is also correct).

b) Find $\text{Ker} T = \{x \in \mathbb{R}^4 : T(x) = 0\}$.

Since T is a matrix map, $\text{Ker} T = \text{Nul} A$, where A is the standard matrix of T . Therefore it suffices to find a basis for $\text{Nul} A$. For that we search for solutions of $Ax = 0$, that is, row reduce $[A : 0]$.

$$\left[\begin{array}{cccc|c} 1 & 4 & 5 & 2 & 0 \\ 2 & 1 & 3 & 0 & 0 \\ -1 & 3 & 2 & 2 & 0 \end{array} \right] \sim \left[\begin{array}{cccc|c} 1 & 0 & 1 & -2/7 & 0 \\ 0 & 1 & 1 & 4/7 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \rightarrow \begin{cases} x_1 + x_3 - \frac{2}{7}x_4 = 0 \\ x_2 + x_3 + \frac{4}{7}x_4 = 0 \end{cases} \begin{cases} x_1 = -x_3 + \frac{2}{7}x_4 \\ x_2 = -x_3 - \frac{4}{7}x_4 \end{cases} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = x_3 \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 2/7 \\ -4/7 \\ 0 \\ 1 \end{bmatrix}$$

so a basis for $\text{Ker} T = \left\{ \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2/7 \\ -4/7 \\ 0 \\ 1 \end{bmatrix} \right\}$.

c) Find the range of T .

Since T is a matrix map, the range of T is the same as $\text{Col} A$. We can find a basis for $\text{Col} A$ by taking the pivot columns of A . From the row reduction in part b) we see the first two columns are pivot columns, therefore a basis for $\text{Col} A$ is $\left\{ \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 4 \\ 1 \\ 3 \end{bmatrix} \right\}$.

② Let $A = \begin{bmatrix} a & a & a & a \\ a & b & b & b \\ a & b & c & c \\ a & b & c & d \end{bmatrix}$, where $a, b, c, d \in \mathbb{R}$.

a) Find the conditions and the LU-factorization of A .

In order to perform an LU-factorization without row interchanges, we need a pivot on every step, so first we need $a \neq 0$. Now proceed:

$$A = \begin{bmatrix} a & a & a & a \\ a & b & b & b \\ a & b & c & c \\ a & b & c & d \end{bmatrix} \sim \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & b-a & c-a & c-a \\ 0 & b-a & c-a & d-a \end{bmatrix} \sim \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & 0 & c-b & c-b \\ 0 & 0 & c-b & d-b \end{bmatrix} \sim \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & 0 & c-b & c-b \\ 0 & 0 & 0 & d-c \end{bmatrix}$$

We need $b-a \neq 0$ We need $c-b \neq 0$ We need $d-c \neq 0$

So in order to be able to do it we need $a, b-a, c-b, d-c \neq 0$ and then $U = \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & 0 & c-b & c-b \\ 0 & 0 & 0 & d-c \end{bmatrix}$ and $L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$.

b) Under such conditions, is A invertible? Why?

Under the previous conditions A has a pivot position in every column (and it's square), so by the Invertible Matrix Theorem, A is invertible.

c) For $a=1, b=2, c=3, d=4$ find A^{-1} .

Since A is invertible we know $[A : I] \sim [I : A^{-1}]$, so

$$[A: I] = \left[\begin{array}{cccc|cccc} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 2 & 2 & 2 & 0 & 1 & 0 & 0 \\ 1 & 2 & 3 & 3 & 0 & 0 & 1 & 0 \\ 1 & 2 & 3 & 4 & 0 & 0 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 2 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 & 1 \end{array} \right] = [I: A^{-1}]$$

③ Let $A_n = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & 2 & 2 & \dots & 2 \\ 1 & 2 & 3 & \dots & 3 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 2 & 3 & \dots & n \end{bmatrix}$ with $n \geq 1$ a generalization of the matrix of the previous exercise. Calculate $\det A_n$.

If we row reduce A_n just like in exercise 2 we will get $A_n \sim \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 0 & 2-1 & 2-1 & \dots & 2-1 \\ 0 & 0 & 3-2 & \dots & 3-2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & n-(n-1) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 0 & 1 & 1 & \dots & 1 \\ 0 & 0 & 1 & \dots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{bmatrix}$, which has determinant 1, since it's a triangular matrix and its determinant is the product of the diagonal entries. Since adding a multiple of one row onto another doesn't change the determinant, we can say $\det A_n = 1$.

④ Let $H = \left\{ \begin{bmatrix} x_1 \\ 0 \\ x_3 \\ 0 \end{bmatrix} : x_1, x_3 \in \mathbb{R} \right\} \subset \mathbb{R}^4$.

a) Show that H is a subspace of $\mathbb{R}^4$.

There are two ways of doing this. First we can see that $x \in H$ means $x = \begin{bmatrix} x_1 \\ 0 \\ x_3 \\ 0 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$, so every element of H is a linear combination of u_1 and u_2 , therefore $H = \text{Span}\{u_1, u_2\}$. Since a spanning set is a vector subspace, we can conclude H is a subspace.

Alternatively it can be proved using the definition.

1- $0 \in H$? Yes! $0 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \in H$.

2- If $u, v \in H$, $u+v \in H$? Let $u = \begin{bmatrix} u_1 \\ 0 \\ u_3 \\ 0 \end{bmatrix}$ and $v = \begin{bmatrix} v_1 \\ 0 \\ v_3 \\ 0 \end{bmatrix}$, then $u+v = \begin{bmatrix} u_1+v_1 \\ 0 \\ u_3+v_3 \\ 0 \end{bmatrix}$ which is in H , since $u_1+v_1, u_3+v_3 \in \mathbb{R}$.

3- If $u \in H$ and $c \in \mathbb{R}$, $cu \in H$? Let u be as before, then $cu = \begin{bmatrix} cu_1 \\ 0 \\ cu_3 \\ 0 \end{bmatrix}$ which is in H , since $cu_1, cu_3 \in \mathbb{R}$.

Since H satisfies all three conditions, it is a subspace of $\mathbb{R}^4$.

b) Find a basis for H .

As explained in a), any vector in H , $\begin{bmatrix} x_1 \\ 0 \\ x_3 \\ 0 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$, so $H = \text{Span}\{u_1, u_2\}$, since $\{u_1, u_2\}$ is linearly independent it is a basis.

c) Let $L = \left\{ \begin{bmatrix} x_1 \\ 1 \\ x_3 \\ 1 \end{bmatrix} : x_1, x_3 \in \mathbb{R} \right\} \subset \mathbb{R}^4$. Is L a subspace? Why?

No, because it doesn't satisfy any of the 3 conditions for it to be a subspace. Even with $x_1 = x_3 = 0 \rightarrow \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}$, so $0 \notin L$.

⑤ Let $A = \begin{bmatrix} 3 & 0 & 0 \\ -3 & 4 & 9 \\ 0 & 0 & 3 \end{bmatrix}$.

a) Find all the eigenvalues of A . Can we know from this if A is diagonalizable?

For that we need to solve $\det(A - \lambda \text{Id}) = 0$: $\begin{vmatrix} 3-\lambda & 0 & 0 \\ -3 & 4-\lambda & 9 \\ 0 & 0 & 3-\lambda \end{vmatrix} = (3-\lambda) \begin{vmatrix} 4-\lambda & 9 \\ 0 & 3-\lambda \end{vmatrix} = (3-\lambda)^2 (4-\lambda) = 0$, so $\lambda_1 = 3, \lambda_2 = 4$ are the eigenvalues.

With this information above we cannot tell whether A is diagonalizable.

b) Give a basis for every eigenspace of A .

Start with $\lambda_1 = 3$. We want to solve $(A - 3I)x = 0$, so row reduce $[A - 3I | 0]$:

$$\left[\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ -3 & 1 & 9 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \text{ which directly gives } -3x_1 + x_2 + 9x_3 = 0 \rightarrow 3x_1 = x_2 + 9x_3 \rightarrow x_1 = \frac{1}{3}x_2 + 3x_3, \text{ so}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} 1/3 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix} = x_2 \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}, \text{ so a basis is } \left\{ \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Now with $\lambda_2 = 4$, same process:

$$\left[\begin{array}{ccc|c} -1 & 0 & 0 & 0 \\ -3 & 0 & 9 & 0 \\ 0 & 0 & -1 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 0 & 9 & 0 \\ 0 & 0 & -1 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \rightarrow \begin{matrix} x_1 = 0 \\ x_3 = 0 \end{matrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \text{ so a basis is } \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

c) Is A diagonalizable? If so give D and P such that $A = PDP^{-1}$.

Yes, it is, since the dimension of each eigenspace matches the multiplicity of its eigenvalue. Thus,

$$D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}, P = \begin{bmatrix} 1 & 3 & 0 \\ 3 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

⑥ Let $W = \text{Span} \left\{ \overset{u_1}{\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}}, \overset{u_2}{\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}}, \overset{u_3}{\begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}} \right\}$

a) Find an orthonormal basis for W .

Clearly $\{u_1, u_2, u_3\}$ are not orthogonal since $u_1 \cdot u_2 \neq 0$, but they are linearly independent by inspection. But we can check: $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix}$, has three pivot columns. So $\{u_1, u_2, u_3\}$ is a basis for W . We can find an orthogonal one by applying the Gram-Schmidt process.

$$v_1 = u_1, \quad v_2 = u_2 - \frac{u_2 \cdot v_1}{v_1 \cdot v_1} v_1 = \begin{bmatrix} -1/2 \\ -1/2 \\ 1/2 \end{bmatrix}, \quad v_3 = u_3 - \frac{u_3 \cdot v_1}{v_1 \cdot v_1} v_1 - \frac{u_3 \cdot v_2}{v_2 \cdot v_2} v_2 = \begin{bmatrix} 1/2 \\ 1/2 \\ -1/2 \end{bmatrix}. \text{ Clearly } \|v_1\| \neq 1, \text{ so in order to get an orthonormal basis, we must normalize.}$$

$$\|v_1\| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3} = \sqrt{3}, \text{ so } v_1' = \frac{1}{\|v_1\|} v_1 = \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}. \text{ Notice that } \|v_2\| = \|v_3\| = 1, \text{ so } \{v_1', v_2, v_3\} \text{ is an orthonormal basis for } W.$$

b) Calculate the distance between $y = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ and W .

Notice that $y = u_2 + u_3$, so $y \in \text{Span}\{u_2, u_3\} \rightarrow y \in \text{Span}\{u_1, u_2, u_3\} = W$, so since y is already in W , the distance is 0. It can also be computed via orthogonal projection.

⑦ True or False?

a) If T is a map between vector spaces and $T(0) = 0$, then T is linear.

False. Let $T(x) = \sin x$. $T(0) = \sin 0 = 0$ but T is not linear because $T(x+y) = \sin(x+y) \neq \sin x + \sin y = T(x) + T(y)$.

b) Let $A_{m \times n}$ and $B_{n \times n}$, then $\det(A+B) = \det A + \det B$.

False. Take $A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$, $A+B = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$ and $\det(A+B) = 25$ but $\det A = 4$ and $\det B = 9$, so $\det A + \det B = 13 \neq \det(A+B)$.

c) Let $A_{4 \times 7}$. It is possible that $\text{rank } A = 5$ and $\text{nullity } A = 2$.

False. A cannot have $\text{rank } 5$ since it only has 4 rows and therefore a maximum of 4 pivot columns.